

1. Global Production Web Application (Vercel Global CDN)

- Main Application:** <https://redzone-x.vercel.app>
Public landing page with 1-click Instant SOS Evacuation Pass (No Login Required).
- Satellite GIS Map:** <https://redzone-x.vercel.app/map>
Esri World 0.3m Real-Time Satellite Map + GPS Turn-by-Turn Navigator + Voice SOS.
- Operations Dashboard:** <https://redzone-x.vercel.app/dashboard>
Multi-hazard vulnerability KPIs, carrying capacity index (CCI), and relief allocations.
- Collaborative Alerts:** <https://redzone-x.vercel.app/alerts>
Live multi-user emergency broadcasts synchronized in real time via MongoDB Atlas Cloud.
- Admin Command Center:** <https://redzone-x.vercel.app/admin>
Automated IMD Radar, GSI sensors & CWC river gauge fact-checking intelligence center.
- Shelters & Sphere Standards** <https://redzone-x.vercel.app/shelters>
Live compliance monitor tracking 45 LPCD drinking water and 3.5 m2/person living space.

2. Backend API & Cloud Database Infrastructure

- Backend API Root:** <https://redzone-x.onrender.com/>
Hosted on Render.com Cloud (Node.js / Express micro-cached REST API).
- Health & Telemetry API:** <https://redzone-x.onrender.com/api/health>
Live health check returning active MongoDB connection status, latency, and jurisdictions.
- Central Alerts API:** <https://redzone-x.onrender.com/api/alerts>
Live database endpoint supporting GET, POST, and DELETE operations with MongoDB Atlas.
- GitHub Repository:** https://github.com/Manojpuchakayala/redzone_x
Full source code repository with automated Vercel & Render CI/CD deployment pipelines.

3. Platform Innovations & SIH Problem Statement Alignment

- Real-Time Satellite Imagery: High-res Esri orbital photography with 3-layer switcher (Satellite, 3D Terrain, Street).
- Turn-by-Turn GPS Navigator: Google Maps-style navigation HUD with Voice SOS audio and detour rerouting.
- Dynamic Road Blockade Switcher: Reroutes convoys via Elevated Ridge Bypass Lines when debris blocks roads.
- Sphere Humanitarian Standards: Guaranteed 45 LPCD drinking water and 3.5 m2/person shelter space quotas.
- Central Cloud Database: MongoDB Atlas (consumer-trust-db.rxdifnq.mongodb.net) with 3-second live sync.

4. Offline Backup & 1-Click Desktop Launcher

- Backup Zip Archive: C:\Users\manoj\Downloads\RED-ZONE-X_Project.zip
- 1-Click Windows Launcher: Unzip and double-click START_PROJECT.bat to launch everything locally.